DEVICEOPS / PROFESSIONAL AGENTS

Autonomous Operations & Incident Resolution

Observe → investigate → authorize → act → verify. Built with the AWS Strands Agents SDK.

01 DEVICE & DELIVERY LAYER

Controlled LAN / local notification services

OBSERVATION SOURCES

Devices & simulators

LAN equipment: read-only probes
Standalone HTTP simulator
Physical/cloud writes disabled

NODE.JS EDGE COLLECTOR

Normalize & deliver

Ping · SNMP system / ports · PJLink
Encrypted vault + durable spool
Outbound API + command journal

↑ simulator-only commands

MOBILE PHOTO CAPTURE

Authenticated upload to v2 API
SHA-256 + incident association

INTEGRATION BOUNDARY

57-entry connector catalog

4 adapters implemented; 53 planned.
Physical/cloud writes disabled.

LOCAL NOTIFICATIONS

n8n → Mailpit

Email delivery + signed callback

telemetry

02 DEVICEOPS APPLICATION

Next.js + TypeScript worker + PostgreSQL

OPERATOR ENTRY POINT

Next.js operations UI + v2 API

Sessions · tenant scope · manager approvals
Collector ingestion · MCP tool endpoint

persisted state

AUTONOMOUS ORCHESTRATION

TypeScript operations worker

EWMA drift: temp, memory, optical loss
PostgreSQL-backed pg-boss investigation jobs
Campus topology & calculated impact

validated proposal

BACKEND OWNS AUTHORIZATION

Policy → execution → verification

Service restart: automatic when policy allows
Switch restart: manager approval + exact hash
One command; 3 fresh healthy samples or escalate

PERSISTENCE

PostgreSQL 17

Tenant RLS · baselines · memory · jobs · audit

investigate

03 AGENT & MODEL LAYER

Private Python service + external model API

CONFIGURED / ACCESS BLOCKED

Google Gemini API

gemini-3.7-flash · external to AWS
Latest auth: HTTP 401; live retest pending

inference

PYTHON / FASTAPI

AWS Strands Agents SDK

Supervisor → specialists → critic → diagnosis
Shared budget + cancellation + scoped evidence
Proposes actions; backend owns authorization

tool calls

STDIO MCP → SCOPED API READS

Scoped evidence & analysis tools

Incident · health · logs · topology
Recent changes · runbooks · capabilities
Short-lived invocation grant; no device secrets

MCP reads use tenant-scoped API

OPTIONAL / ACCOUNT-BLOCKED

Amazon Bedrock + Titan

Nova reasoning; Titan semantic runbooks

RETRIEVAL Gemini uses lexical runbook search. Optional Titan vectors are evaluated in the service; no hosted vector database is claimed.

END-TO-END INCIDENT LIFECYCLE

Observe → Detect & group → Investigate → Authorize → Execute → Verify → Report / notify

MULTIMODAL EVIDENCE

Quarantine → scan → isolated parser

Mobile photo & manual PDF (>pg 5 citations)
ClamAV malware scan + SHA-256 integrity
Device associations require confirmation
Only clean, approved artifacts enter context

SANDBOXED ANALYSIS

Private broker → disposable container

No network · credentials · application mounts
1 CPU · 256 MiB · 64 PIDs · 30 seconds
Telemetry CSV + synthetic waveform FFT
Measured spectra; no bearing-fault diagnosis

ORGANIZATIONAL MEMORY

Incident / knowledge / fleet memory

Factual outcomes + reviewed runbooks
Source IDs, versions and tenant isolation
EWMA baselines + explicit evidence gaps
AgentCore: separately unverified

VERIFICATION SNAPSHOT · 10 SEP 2026

Current demo: fixture reasoning. Live specialist inference, phone camera and AWS adapters need separate verification.